Assignment Report – Change Detection & Object Removal

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Assignment: 01

1. Grayscale images ploting

2. Background Modeling

We start by constructing a background model using the **mean** and **variance** of the first set of frames.

- For *person1*, first **70 frames** are used.
- For *person3*, first **60 frames** are used.

- For other datasets, all frames are used.

Results:

- Figure: Mean Image (represents background).
- Figure: Variance Image (represents variability of each pixel).

Mean | Variance

3. Change Detection (Mahalanobis Distance)

To detect motion, we compute the Mahalanobis distance between each frame and the background model. Pixels above a threshold are marked as foreground (motion).

We experimented with **different thresholds**:

- **Threshold = 1:** Detects even small differences (many false positives).
- **Threshold = 3:** Balanced detection (good separation).
- **Threshold = 5:** Only large changes are detected (small movements missed).

Results:

- Figure: Masks for threshold = 1, 3, 5 (side by side comparison).

Threshold=1 Threshold=3 Threshold=5

4. Noise Removal (Morphological Operations)

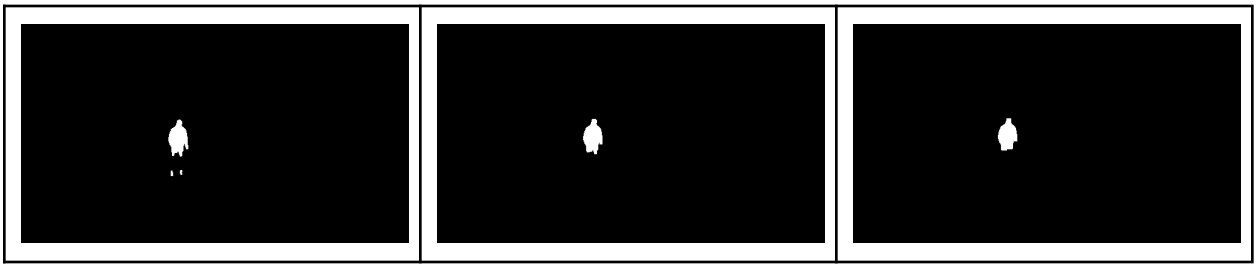
The raw masks contain noise. We apply morphological **opening** (erosion → dilation) with different kernel sizes:

- **Kernel 3×3:** Removes small noise but keeps object shape.
- **Kernel 5×5:** More aggressive cleaning, may shrink object.
- **Kernel 7×7:** Strong cleaning, small objects disappear.

Results:

- Figure: Morphological masks for 3×3, 5×5, 7×7 kernels.

3x3 5x5 7x7



5. Connected Components Analysis

After cleaning, we perform connected components labeling:

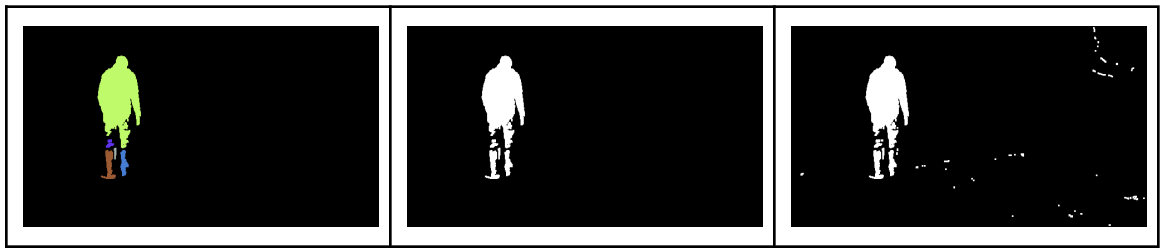
- Each blob (object in motion) is detected.
- For each blob, we compute:
    - **Area (size in pixels).**
    - **Centroid (average position).**
    - **Bounding Box (object boundaries).**

Small blobs are filtered out as noise.

Results:

- Figure: Binary cleaned mask.
- Figure: Connected components visualization (colored blobs).

Labeled figure Without labeled mask image.



6. Object Removal (Alpha Blending)

For person datasets, we remove the moving object by replacing it gradually with the background using **alpha blending**.

- Alpha increases **gradually** from 0 → 1 across frames.
- At the start, object is visible; at the end, object is replaced by background.

Results:

- Figure: Sample alpha-blended frames (early, middle, final).

7. Videos and Outputs

All intermediate results are saved as both images and videos:

- **Mean and Variance images.**
- **Original frames (grayscale).**
- **Motion masks.**
- **Morphological masks.**
- **Connected component masks and labels.**
- **Alpha-blended frames.**
- **Videos (.mp4) of each stage.**

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8. Analysis & Conclusion

- Lower thresholds (1) detect more motion but add noise.

- Higher thresholds (5) suppress noise but may miss small motions.
- Kernel size **3×3** gave the best trade-off between noise removal and object preservation.
- Connected components helped isolate true moving objects and reject noise.
- Alpha blending successfully removed the moving person from the video background.